

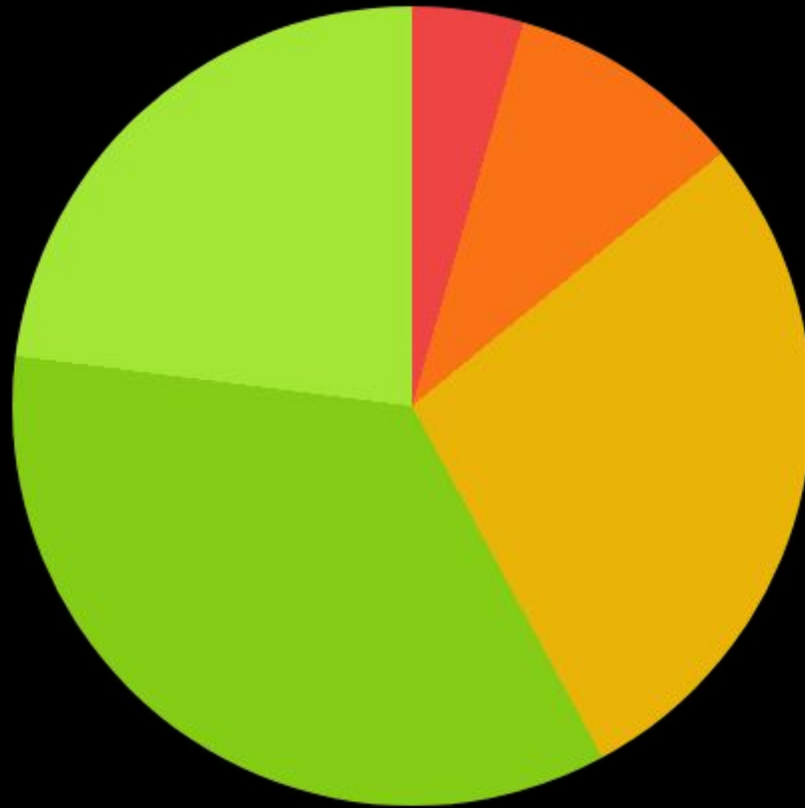
Personalized Recommendation Engine

Benchmarking Latent Factor Optimization vs. Neighborhood Filtering on the
Netflix Prize Dataset

The Discovery Challenge

Scaling content discovery across 2M+ interaction samples while overcoming structural data sparsity.

User Sentiment Landscapes



- 1.0 Star: 89,009 (4.5%)
- 2.0 Stars: 192,943 (9.6%)
- 3.0 Stars: 559,630 (28.0%)
- 4.0 Stars: 697,744 (34.9%)
- 5.0 Stars: 460,674 (23.0%)

| Dual Algorithmic Methodology



SVD Latent Factors

Singular Value Decomposition projects users and items into a 15-dimensional latent space to uncover hidden preference vectors.



Item-Based Neighborhood

Collaborative Filtering using Cosine Similarity between movie rating patterns in a memory-based memory structure.

Structural Sparsity Thresholds

98.4%

Matrix Density Gap

Complexity Metrics

The extreme sparsity of the interaction matrix confirms that memory-based systems face severe connectivity bottlenecks.

- Unique Users Sampled: 342,445
- Avg Ratings per User: 5.84
- Avg Ratings per Movie: 5540.17

Model Performance Benchmarks

SVD (MAP@10)

0.9209

KNN (MAP@10)

0.9037

SVD (RMSE)

0.9852

KNN (RMSE)

1.0906

Varied Personalized Execution

Target Profile	Top-Ranked Personalized Discovery	Score	Logic Applied
NEW_USER_999	Futurama: Monster Robot Maniac Fun...	4.302	Cold-Start Fallback
User 1434636	Invader Zim	4.048	Personalized (SVD)
User 2442	Aqua Teen Hunger Force: Vol. 1	4.427	Personalized (SVD)
User 1196100	Star Trek: Deep Space Nine: Season 5	4.481	Personalized (SVD)
User 2207774	Lord of the Rings: Return of the King...	4.865	Personalized (SVD)

Execution Note: Discovery paths vary based on latent genre affinity, ranging from Sci-Fi cult classics to epic fantasy.

Strategic Roadmap

Future Optimization

Deploying model-based SVD provides superior storage efficiency and faster real-time inference across diverse catalogs.

Future iterations will leverage content-based metadata (genres, cast) to refine long-tail discovery and enhance cold-start accuracy.

Integration of Neural Collaborative Filtering (NCF) will be explored to capture higher-order non-linear user-item interactions.

